

JULAIA PORT, Kuwait

WMO: 405930

Lat: 28.867N

Lon: 48.283E

Elev: 29

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 07-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	45.3	48.0	22.7	17.1	50.0	26.6	20.5	52.7	47.5	62.9	43.7	59.5	13.2	300	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	105.0	74.3	103.4	74.7	101.8	75.3	88.6	94.4	87.4	94.1	86.3	94.1	17.6	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
87.1	198.0	93.7	85.8	189.5	93.4	84.4	180.7	93.2	53.3	94.6	52.0	94.0	50.6	93.8	98.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
41.1	34.2	28.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.5	57.4	60.2	67.3	75.9	86.3	94.3	96.6	96.0	91.9	83.6	70.4	61.1
	DBStd	14.80	5.15	4.70	4.23	4.26	4.17	3.04	3.01	2.71	2.73	3.37	5.15	4.78
	HDD50	8	7	1	0	0	0	0	0	0	0	0	0	0
	HDD65	552	241	140	24	0	0	0	0	0	0	0	12	135
	CDD50	10419	235	288	535	777	1126	1329	1445	1427	1257	1042	613	345
	CDD65	5488	4	7	94	328	661	879	980	962	807	577	175	14
	CDH74	80245	17	1	133	1958	8931	14524	16635	16371	13193	7582	896	4
	CDH80	51765	8	0	27	324	4626	10204	12171	11907	8875	3508	115	0
(14) Wind	WSAvg	12.9	13.5	13.7	13.9	11.0	12.9	14.7	12.6	12.1	11.3	12.2	13.7	13.0
(15) Precipitation	PrecAvg	7.2	1.4	0.9	1.2	0.7	0.2	0.0	0.0	0.0	0.0	0.2	1.3	1.2
	PrecMax	11.7	4.1	3.2	3.8	2.0	0.8	0.1	0.1	0.1	0.0	1.0	7.2	3.5
	PrecMin	3.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.4	1.0	0.8	0.9	0.6	0.2	0.0	0.0	0.0	0.0	0.2	1.5	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.2	71.5	83.9	90.0	99.2	105.9	107.7	106.9	103.3	96.6	85.3	73.5
		MCWB	81.9	62.0	62.9	66.3	70.7	72.3	73.0	76.2	75.8	74.7	70.0	63.0
	2%	DB	68.3	69.3	76.8	85.6	95.7	103.3	105.3	104.3	100.7	93.6	82.9	71.7
		MCWB	61.5	60.8	61.7	66.8	71.8	73.1	73.8	75.8	75.6	72.7	70.3	63.5
	5%	DB	66.5	67.8	74.5	83.3	93.8	101.4	103.7	102.7	99.1	91.4	81.0	70.4
		MCWB	59.9	59.9	63.2	67.4	72.0	73.7	74.1	76.6	75.9	74.5	68.7	62.5
	10%	DB	64.8	66.6	72.8	81.5	92.1	99.8	102.1	101.1	97.6	90.0	78.8	69.2
		MCWB	58.6	59.5	62.8	67.6	72.5	73.8	74.7	77.1	76.6	74.2	66.9	61.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.7	65.6	68.0	73.8	80.9	85.4	89.5	90.1	88.3	83.7	76.7	68.1
		MCDB	70.6	68.3	72.6	82.1	89.3	94.0	94.1	95.2	94.1	89.9	82.3	71.7
	2%	WB	63.7	63.7	67.1	72.6	79.5	83.6	87.7	89.1	86.6	81.7	73.9	66.3
		MCDB	66.5	66.9	71.9	80.6	88.6	93.8	94.0	94.7	93.6	88.9	80.1	70.4
	5%	WB	61.6	62.5	66.3	71.4	78.2	82.1	86.3	88.1	85.6	80.0	71.1	64.4
		MCDB	65.0	66.3	71.3	79.2	88.0	93.7	93.7	94.3	93.3	87.3	78.5	69.3
	10%	WB	59.6	61.1	65.5	70.3	77.0	80.3	85.1	87.2	84.0	78.6	69.0	62.5
		MCDB	63.9	65.5	70.7	78.1	87.5	94.3	93.9	94.4	93.1	86.3	76.6	67.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	8.7	8.1	8.1	8.0	10.0	10.1	9.3	10.6	10.7	10.9	10.9
		MCDBR	9.4	8.5	10.3	11.0	10.5	12.4	12.4	11.7	12.9	12.2	12.4	11.0
		MCWBR	8.8	8.5	9.7	10.7	12.9	13.7	14.3	14.6	15.9	14.7	11.8	10.1
	5% WB	MCDBR	8.2	6.3	6.3	8.3	8.0	9.6	8.2	8.0	9.2	9.9	9.7	8.2
		MCWBR	8.9	7.4	7.4	9.3	12.1	16.8	14.6	13.2	13.8	13.0	10.3	9.2
(40) Clear-Sky Solar Irradiance	taub		0.516	0.564	0.617	0.703	0.703	0.663	0.681	0.631	0.576	0.569	0.550	0.513
	taud		1.823	1.729	1.617	1.519	1.525	1.567	1.567	1.657	1.751	1.764	1.793	1.842
	Ebn at Noon		219	221	220	207	208	216	211	220	227	217	207	212
	Edh at Noon		57	68	81	92	92	87	87	79	70	65	58	54
(44) All-Sky Solar Radiation	RadAvg		1072	1325	1660	1812	2084	2310	2207	2110	1895	1522	1115	983
	RadStd		73	87	83	99	92	88	59	48	67	63	85	89

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.59	N/A	N/A	N/A	+0.97	N/A	N/A	N/A	N/A	+169	+193

Nomenclature: See separate page